# 10-Mark Questions & Answers

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Q1. Explain CPU architecture in detail with a block diagram - covering ALU, Control Unit, Registers, Cache Memory, and Bus System with their functions.

## 1. Comprehensive CPU Block Diagram

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```

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## 2. Functional Unit Details

1. Control Unit (CU):

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Q2. Explain all four types of instruction formats (three-address, two-address, one-address, zero-address) in detail with structure, examples, advantages, and disadvantages.

## 1. Format Structures

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## 2. Program Execution Comparison for `X = (A + B) * (C + D)`

#### A. Three-Address Instruction Format

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```

#### B. Two-Address Instruction Format

```
text
```

#### C. One-Address Instruction Format (Accumulator Architecture)

```
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```

#### D. Zero-Address Instruction Format (Stack Architecture)

```
text
```

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Q3. Explain the basic instruction cycle in detail with register transfer operations, and explain interrupt processing with a diagram.

## 1. Instruction Cycle Stages & Register Transfer Notation

1. Fetch Stage:

* `MAR

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## 2. Interrupt Processing Cycle

```
text
```

Normal

* **Step-by-step:**

1. CPU

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Q4. Explain microprogrammed control unit in detail - block diagram, working for ADD AX, BX, microinstruction format, pros/cons, and comparison with hardwired.

### 1. Block Diagram

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### 2. Microinstruction Format

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### 3. Step-by-step Execution for `ADD AX, BX`

 * **Address 100:** Control Signals = `Temp1 <- AX`; Next Address = `101`.

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### 4. Comparison Table

| Feature | Hardwired Control Unit | Microprogrammed Control Unit |

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Q5. Explain hardwired control unit in detail - block diagram, working for ADD AX, BX, example of control signal generation, pros and cons.

## 1. Detailed Block Diagram

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```

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## 2. Control Signal Generation Example

Consider control signal `Ld_AX` (Load AX register):

* `Ld_

at state

Equation:

`Ld\_AX`

* **Hardware Implementation:** A 3-input OR gate driven by three AND gates.

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## 3. Pros and Cons

* **Pros:** High speed (operates at native clock rate); compact footprint for RISC.

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Q6. Explain microinstruction sequencing and execution in detail with block diagram, microinstruction format, and a complete worked example for ADD AX, BX.

## 1. Sequencing Architecture Diagram

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## 2. Complete Worked Example for `ADD AX, BX`

Assume Opcode of `ADD AX, BX` maps to Control Memory Address `200`:

| Control Addr | Micro-Operation Signals | Condition Field (CD) | Next Addr / Sequencing |

| :--- | :-

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Q7. Explain parallel processing concepts in detail - sequential vs parallel, why needed, applications, bit-level and instruction-level parallelism.

## 1. Sequential vs. Parallel Processing

* **Sequential Processing:** Executes one instruction at a time on a single CPU core. Execution time equals sum of all instruction times.

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## 2. Why Parallel Processing is Needed

1. Physical Power Wall: Transistor clock rates capped at ~5 GHz due to extreme heat dissipation.

2. High

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## 3. Levels of Parallelism

```
text
```

BIT-LE

INSTRUCTION-LEVEL PARALLELISM (ILP):

Non-Pi

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Q8. Explain Flynn's classification in detail with the classification matrix, diagrams, examples, real-life analogies, and applications for SISD, SIMD, MISD, and MIMD.

## 1. Flynn's 2x2 Matrix

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```

SINGLE

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## 2. Complete Breakdown of the 4 Architectures

1. SISD (Single Instruction, Single Data):

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Q9. Explain instruction pipelining and pipeline hazards in detail - 5-stage pipeline, pipeline timing diagram, hazards, and hazard-handling techniques.

## 1. 5-Stage Pipeline Execution

```
text
```

Clock C

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## 2. Pipeline Hazards & Solutions

1. Structural Hazard: Two stages need same hardware resource.

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Q10. Draw and explain the pipeline timing diagram for 4 instructions in a 5-stage pipeline; identify a RAW hazard scenario, show how stalling resolves it, and calculate stall cycles.

## 1. RAW Hazard Scenario

Consider the code:

```
text
```

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## 2. Pipeline Timing Diagram WITH STALLING (Resolving RAW Hazard)

```
text
```

Clock C

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## 3. Stall Cycles Calculation

* Without Forwarding: **2 Stall Cycles** are introduced to allow `I1` to write `R1` in `C5` so `I2` can safely read it in `C6`.

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